

Qatar-2b

R-1777

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_220410 · created 2026-08-02T22:50:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459678.4039 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.067
Transit depth (Rp/Rs)^2	3.21 +/- 2.39 [%]
Orbital Inclination (inc)	87.39 +/- 1.76
Airmass coefficient 1 (a1)	18.32 +/- 8.34
Airmass coefficient 2 (a2)	-1.1 +/- 0.83
Scatter in the residuals of the lightcurve fit is	30.69 %
Best Comparison Star	#2 - [455, 190]
Optimal Aperture	1.82
Optimal Annulus	6.75
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.067 vs 0.16327 (deviation 0.23 σ)
Duration fit vs published	0.0 d vs 0.0767 d (deviation inf σ)
Mid-time O-C	-1926.2 min
Depth SNR	2.7
Residual scatter	30.69 %

Light curve

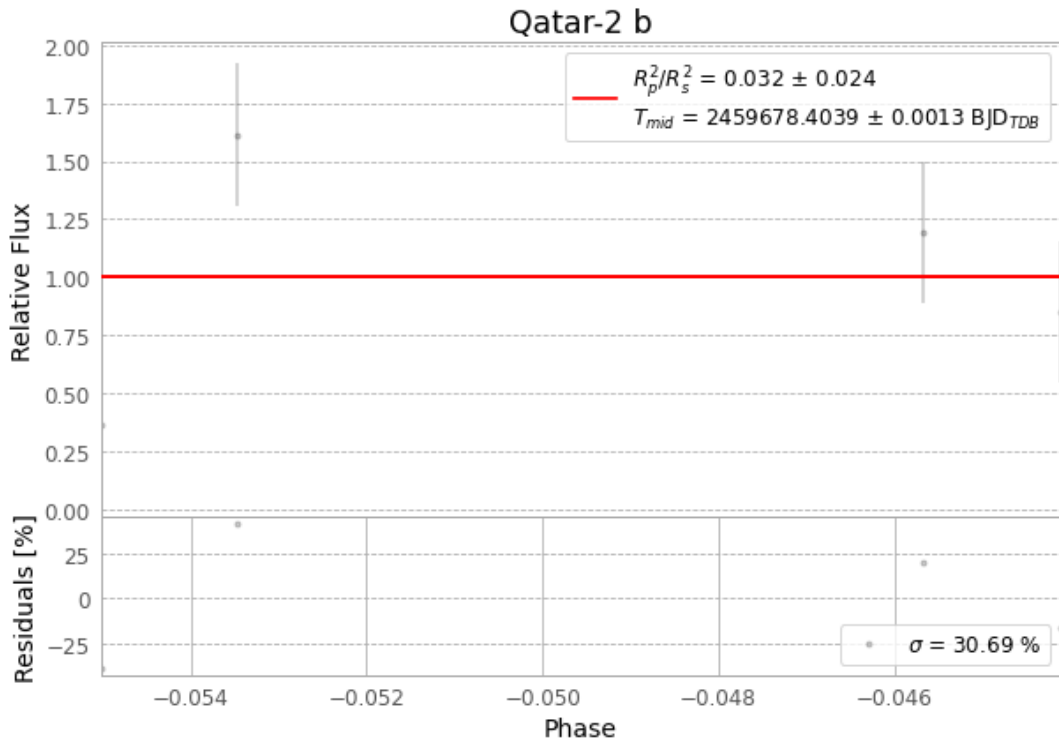


Figure 1 — FinalLightCurve_Qatar-2 b_2022-04-09.png (SHA-256 dafb9106cf2f374d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 23], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7fdefbab4a8e93d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), Qatar-2220410034513.FITS → Qatar-2220410065713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-220410.log (64436819108c...), FinalLightCurve_Qatar-2 b_2022-04-09.png (dafb9106cf2f...), FinalLightCurve_Qatar-2 b_2022-04-09.csv (0fb945ae12e7...)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:50Z.